

```

ReferenceError: stage1 is not defined
aug26db> db.emp.aggregate([
...   {$match: {$job: {$ne: "PRESIDENT"}}},
...
...   {$group: {
...     _id: "$job",
...     totalSal: {$sum: "$sal"},
...     maxSal: {$max: "$sal"},
...     minSal: {$min: "$sal"},
...     avgSal: {$avg: "$sal"}
...   }}
... ])
[
  {
    _id: 'MANAGER',
    totalSal: 8375,
    maxSal: 2975,
    minSal: 2550,
    avgSal: 2791.6666666666665
  },
  {
    _id: 'ANALYST',
    totalSal: 6000,
    maxSal: 3000,
    minSal: 3000,
    avgSal: 3000
  },
  {
    _id: 'SALESMAN',
    localField: "deptno",
    foreignField: "_id",
    as: "department"
  }
],
{$unwind: "$department"},
{
  $project: {
    _id: 0,
    ename: 1,
    dname: "$department.dname",
    income: {$add: ["$sal", "$comm"]}
  }
},
{$sort: {income: 1}}
])
ename: 'SMITH', dname: 'RESEARCH', income: null },
ename: 'JONES', dname: 'RESEARCH', income: null },
ename: 'BLAKE', dname: 'SALES', income: null },
ename: 'CLARK', dname: 'ACCOUNTING', income: null },
ename: 'SCOTT', dname: 'RESEARCH', income: null },
ename: 'KING', dname: 'ACCOUNTING', income: null },
ename: 'ADAMS', dname: 'RESEARCH', income: null },
ename: 'JAMES', dname: 'SALES', income: null },
ename: 'FORD', dname: 'RESEARCH', income: null },
ename: 'MILLER', dname: 'ACCOUNTING', income: null },

```

```

ig26db> db.emp.aggregate([
{
  $lookup: {
    from: "dept",
    localField: "deptno",
    foreignField: "_id",
    as: "department"
  }
},
{$unwind: "$department"},
{
  $project: {
    _id: 0,
    ename: 1,
    dname: "$department.dname",
    income: {
      $add: [
        "$sal",
        ${ifNull: ["$comm", 0]}
      ]
    }
  }
},
{$sort: {income: 1}}
])

    dname: "$department.dname",
    income: {
      $add: [
        "$sal",
        ${ifNull: ["$comm", 0]}
      ]
    }
  }
},
{$sort: {income: 1}}
])

{ ename: 'SMITH', dname: 'RESEARCH', income: 850 },
{ ename: 'JAMES', dname: 'SALES', income: 950 },
{ ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
{ ename: 'MILLER', dname: 'ACCOUNTING', income: 1400 },
{ ename: 'TURNER', dname: 'SALES', income: 1500 },
{ ename: 'WARD', dname: 'SALES', income: 1750 },
{ ename: 'ALLEN', dname: 'SALES', income: 1900 },
{ ename: 'CLARK', dname: 'ACCOUNTING', income: 2550 },
{ ename: 'MARTIN', dname: 'SALES', income: 2650 },
{ ename: 'BLAKE', dname: 'SALES', income: 2850 },
{ ename: 'JONES', dname: 'RESEARCH', income: 2975 },
{ ename: 'SCOTT', dname: 'RESEARCH', income: 3000 },
{ ename: 'FORD', dname: 'RESEARCH', income: 3000 },
{ ename: 'KING', dname: 'ACCOUNTING', income: 5100 }

```